A Fusion of Handcrafted Feature-Based and Deep Learning Classifiers for Heart Murmur Detection

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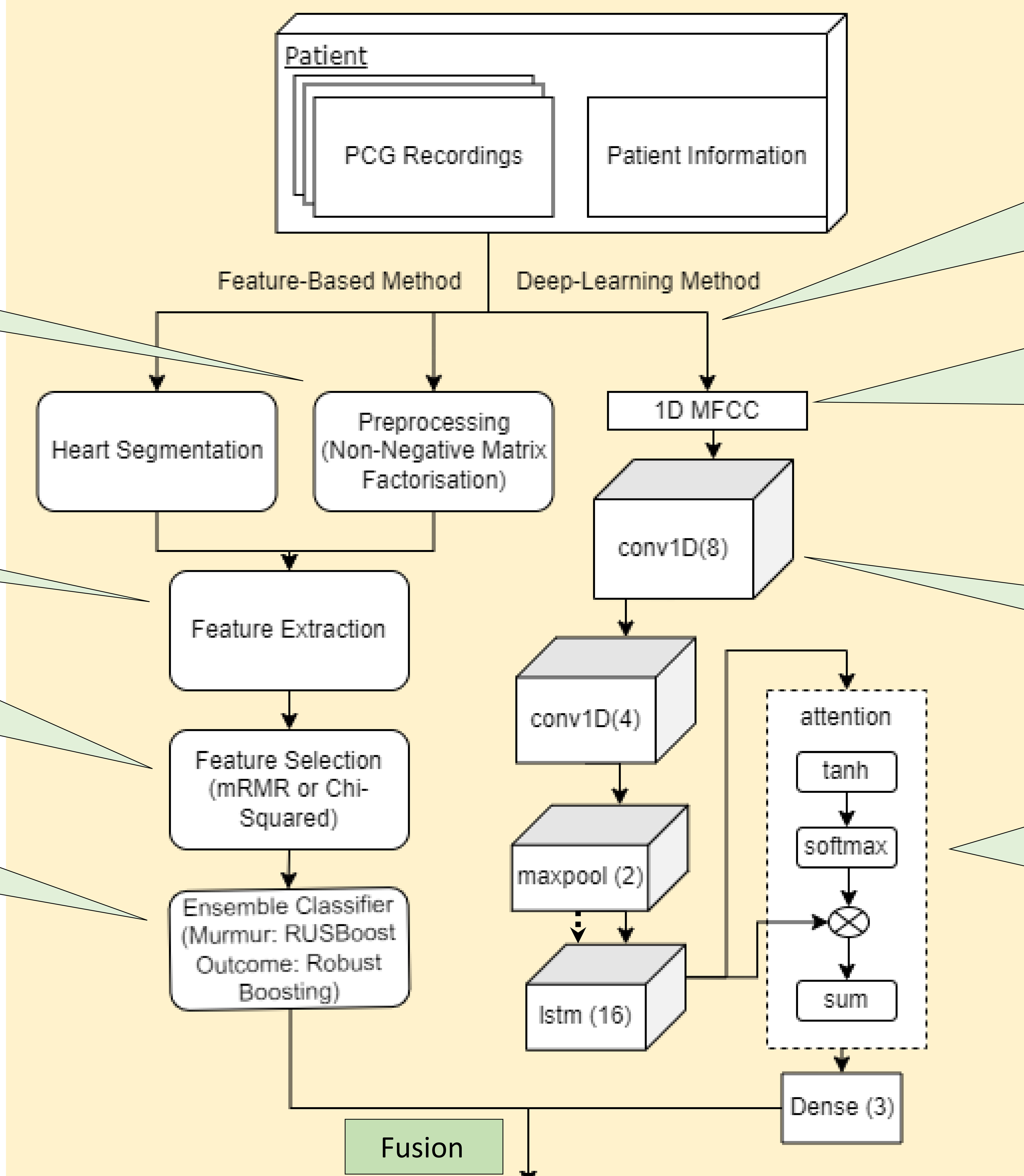
Background: Heart murmur and outcome classification as part of George B. Moody Physionet Challenge [1]
Aim: Investigate fusion of feature-based and deep-learning classifiers on the **Circor Digiscope Phonocardiogram Dataset** [2]

Feature-Based Method

- ❖ Denoising: Non-negative matrix factorisation [4] with clean reference sounds
- ❖ Statistical, Entropy, Power Features, MFCCs, Autocorrelation features extracted from segmented and whole PCGs
- ❖ Feature Selection: maximum Relevance Minimum Redundancy with mutual information quotient or Chi-square test
- ❖ Classification Model: RUSBoost classifier or robust boosting classifier



Voting Methods	
Majority Voting	The most frequent label is selected.
Mean Probabilistic Maximum	The average probabilistic output is used.
Conditional Mean Probabilistic Maximum	Condition checked. Otherwise, average probabilistic output is used.

Deep Learning Method

- ❖ Other classes up/down-sampled to match amount of Murmur Present patients
- ❖ 1D input: 25 MFCC features from ten-second segments of PCG audio averaged in temporal dimension [3]
- ❖ CNN-LSTM with ReLU activation functions

Optional:
- ❖ Bidirectional LSTM layer
- ❖ Attention mechanism

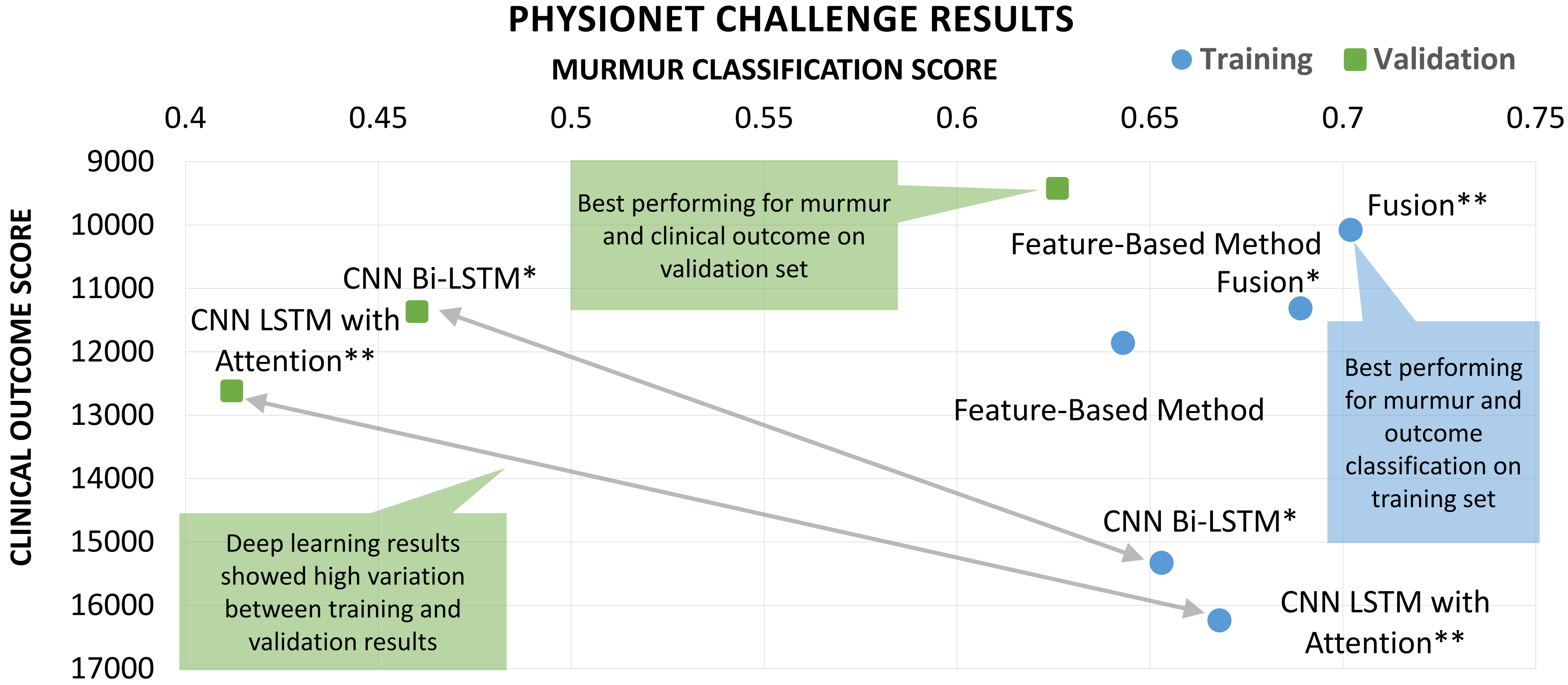
Discussion

- ❖ Murmur Classification:
  - ❖ Performed best with **RUSBoost** due to addressing significant class imbalance
- ❖ Outcome Classification:
  - ❖ Performed best with **robust boosting** avoiding over-concentration on misclassified observations
- ❖ Voting Method:
  - ❖ **Conditional mean probability** performed best as competition cost function places strong weight on correctly identifying abnormal over the normal class

Discussion

- ❖ Classification:
  - ❖ May benefit from **more complex model** or input of 2D MFCC (include temporal dimension of PCG).
- ❖ Voting Method:
  - ❖ Performed best with **Mean Probabilistic Output** due to number of segments

PHYSIONET CHALLENGE RESULTS



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